Innovation in Engineering

Progress Report 3

Goal: The purpose of prioritizing your focus areas is to narrow them down to only the most important ones, before moving on to the next step. Additionally, mapping innovation opportunities helps identify where you could potentially add value for one or more of the stakeholders relevant to your innovation challenge.

a) What challenges did you face when prioritizing focus areas with your team, and why do you think these challenges arose?

b) Should opportunities always be selected based on high value and low complexity? Explain your reasoning.

c) What difficulties did you encounter while mapping opportunities for innovation?

Consider: Team dynamics, time management, opportunity assessment, and risk assessment.

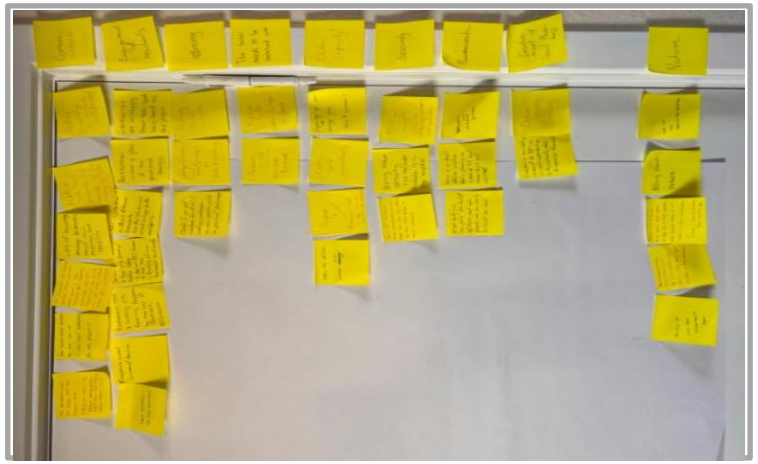
Student name: _Jiayue Che_

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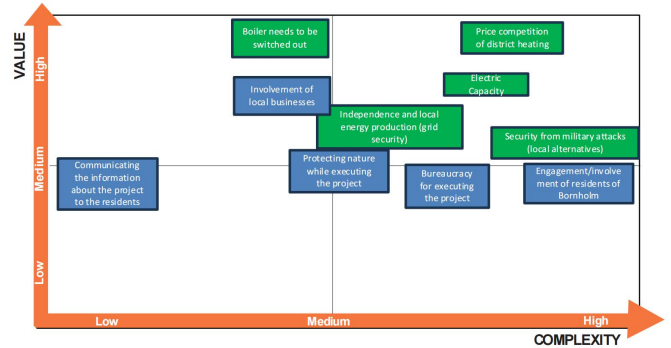
a) One challenge was deciding which focus area was the most important. At the beginning, we had many different opportunities, such as electric capacity, district heating prices, local energy production, energy security, and involvement of residents and local businesses. They were all related to the energy transition on Bornholm, so it was difficult to decide which one we should focus on.
We also had different opinions in the group because everyone looked at the problem from a different perspective. Another challenge was that our understanding of the problem changed when we collected more information. At first, we mainly focused on district heating, but after interviewing different stakeholders, we realized that the limited electricity capacity was actually one of the main problems. So we needed to discuss our focus again and make sure it was supported by the information we collected.

b) No, I don't think so. High value and low complexity are useful when we want to compare many opportunities and quickly find some good options. We also used these two factors in our opportunity evaluation matrix. However, I don't think they should be the only things we consider.
Some opportunities may have high complexity but still be very important in the long term. For example, increasing local electricity capacity on Bornholm may require more investment, infrastructure, and planning, but it could support BEOF's transition from biomass to electricity and also help local businesses. Therefore, we should also consider long-term benefits, energy security, environmental impact, cost, scalability, and the needs of different stakeholders.

c) One difficulty was that many of our opportunities were connected to each other. For example, electricity capacity, local renewable energy production, district heating prices, and energy security are not completely separate topics. Improving one area could also influence the others, so sometimes it was difficult to decide how to separate and compare them.
Another difficulty was that we did not have enough information about every opportunity. It was difficult to know the exact cost, environmental impact, technical complexity, or whether local residents would accept some solutions. We also found that our understanding changed after talking to stakeholders. The interviews helped us realize that the limited electricity capacity was more important than we first thought. This showed us that mapping opportunities is not always a simple or fixed process. We sometimes needed to go back, discuss again, and change our priorities when we got new information.



Template: Opportunity Evaluation Matrix



Goal: the purpose of evaluating a list of opportunities is to assess them from different perspectives and determine which one has the highest potential value.

a) Reflect on your experience arranging the mindset model perspectives from easiest to most challenging

b) What was the most challenging aspect of writing your innovation brief? Why?

c) How did your team find transitioning from generating ideas to evaluating and refining concepts?

Consider: Perspective organization, communication skills, team dynamic.

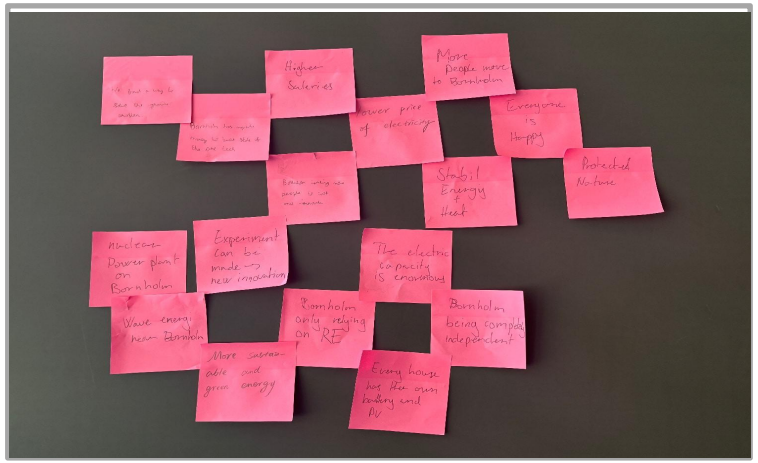
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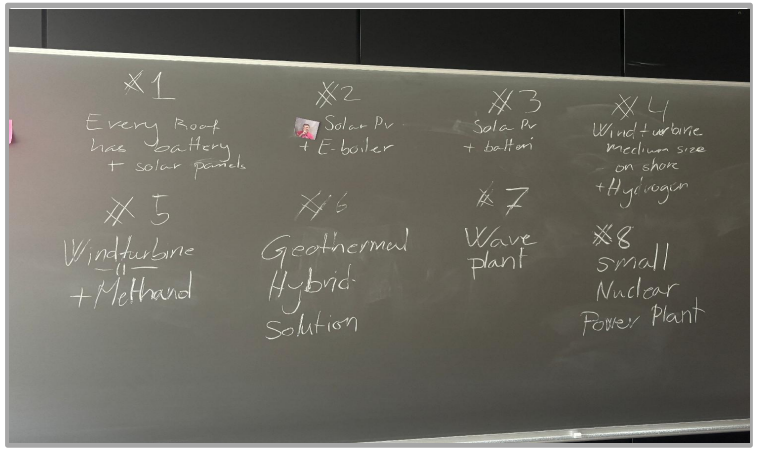
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a) For our team, the technical perspective was relatively easy because as engineering students, we naturally focused on the energy system and technical problems first. For example, we looked at the current heating system, electricity capacity, and different energy solutions on Bornholm.
The economic and environmental perspectives were a little more challenging because we needed more information about costs, investment risks, and environmental impacts. The social and community perspectives were even more difficult because different stakeholders had different needs and opinions. For example, BEOF, local businesses, the municipality, and residents may have different priorities. Residents may also have different opinions about land use and renewable energy projects.
The most challenging part was therefore trying to understand the problem from perspectives that were different from our own engineering background. The Expanded Mental Model helped us realize that the challenge was not only technical. We also needed to think about economic, environmental, social, and ethical aspects. Discussing these perspectives as a team helped us get a broader understanding of the challenge.

b) The most challenging part was identifying and clearly describing our main opportunity. At the beginning, we were mainly looking at district heating and different energy-related problems on Bornholm. However, after our interviews and discussions, we realized that limited electricity capacity was actually one of the main barriers behind several of these problems.
It was also challenging to connect all the information from different stakeholders and explain why this opportunity was important. We needed to consider BEOF, local residents, businesses and the municipality. Finally, we decided to focus on increasing local electricity capacity because it could support BEOF's transition from biomass to electricity and also create value for other stakeholders on Bornholm.

c) The transition was not always easy because during the idea generation stage, we tried to be open and come up with many different ideas without judging them too much. When we moved to the evaluation stage, we had to become more critical and realistic.
We first compared different opportunities based on value and complexity. Later, we generated different energy solutions, such as solar PV, battery storage, wind energy, geothermal energy and wave energy. We then compared them using different criteria, such as cost, environmental impact, energy security and resident acceptance.
Sometimes it was difficult to agree because each idea had different advantages and disadvantages. However, using the evaluation matrix and clear criteria made our discussion easier. This made it easier for us to explain why one idea was better than another. We also combined some ideas and refined them into more realistic concepts. Overall, the evaluation process helped us move from many broad ideas to a smaller number of more realistic and useful solutions.





Goal: The purpose is to reflect on your early-stage concept development process, the importance of quantitative assessment, and the comprehensive evaluation of your final concepts.

a) What steps did your team take to verify that each concept fits within its intended use-context, and how did you identify and address strengths, weaknesses, and needed improvements?

b) Why is it relevant to reassess and update your set of criteria during the concept assessment process, and how did this impact your concept evaluation?

c) How would you evaluate your final concept's potential in relation to the criteria in the mindset model (feasibility, viability, sustainability, and responsibility)?

Consider: Contextual analysis, critical thinking, evaluation process.

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a) We first checked whether our concept fits the situation in Bornholm and especially Nexø. We looked at the local electricity capacity, district heating system, available space, energy demand, and the needs of different stakeholders. We also used the information from our previous interviews to make sure that the concept was related to the real problems on Bornholm.

Then, we listed the important information that we still did not know and turned it into assumptions. For example, we made assumptions about solar energy production throughout the year, available land and rooftop area, existing storage capacity, energy demand, economic feasibility, and residents' willingness to have solar panels.

For each assumption, we also considered what could happen if it was wrong. This helped us identify weaknesses and possible improvements. For example, if solar production is too low in winter, we may need additional energy sources. If the available area is smaller than expected, we may need to reduce the system size or find another location.

b) It is important to reassess the criteria because our understanding of the concept changes when we get more information. At an earlier stage, we mainly compared different opportunities based on value and complexity. However, when we developed our concept further, we realized that we also needed to consider more specific factors.

For example, for our solar and storage concept, solar production, available space, storage capacity, energy demand, cost, energy losses, and resident acceptance are all important. Adding these factors helped us evaluate the concept in a more realistic way instead of only looking at its general value and complexity.

It also helped us identify which assumptions are the most important to test. If some important assumptions are wrong, such as the available space or actual energy demand, the size, cost, or even feasibility of our concept may change.

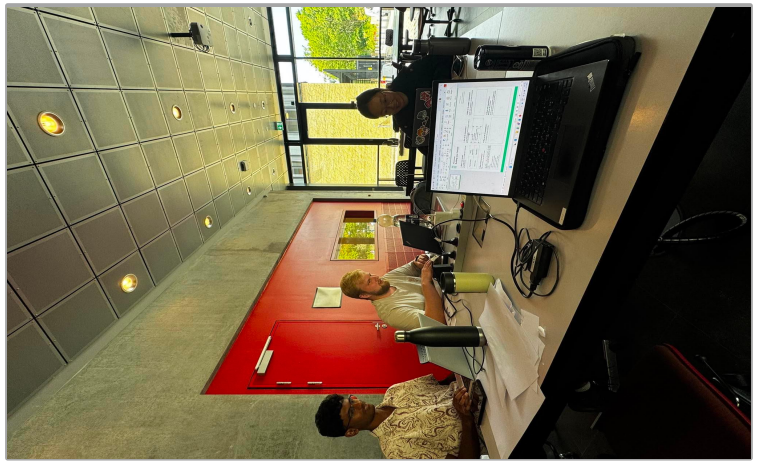
c) Overall, we think our concept has good potential, but some important assumptions still need to be tested.

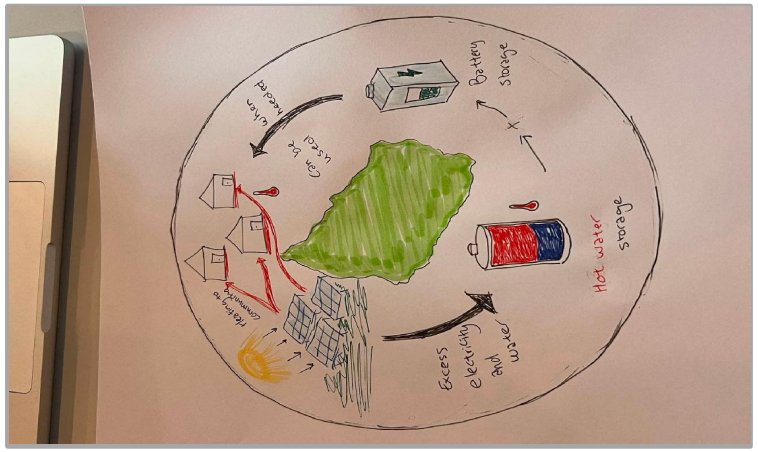
In terms of feasibility, solar panels, batteries, heat storage, and electric boilers are existing technologies, so the concept is technically possible. However, we still need to confirm whether there is enough space, storage capacity, and solar production to meet the energy demand, especially during winter.

For viability, the concept could reduce dependence on imported biomass and use locally produced renewable electricity. However, the investment cost, maintenance cost, and required system size still need to be evaluated.

For sustainability, the concept supports Bornholm's transition away from biomass and increases the use of local renewable energy. Battery and heat storage can also help use excess solar energy instead of wasting it.

For responsibility, we need to consider local residents and other stakeholders. Some residents may not want solar panels on their houses or may be concerned about land use. Therefore, their acceptance and involvement should be considered when developing the concept further.





Goal: The purpose of the first tasks is to reflect on the type of concept you have developed and to choose an appropriate prototype to test if your most crucial assumptions/hypotheses are true or false.

a) How do you plan to use the prototyping results to inform future iterations of your concept and how did its activities impact your priorities or approach moving forward?

b) What considerations are important when prototyping, and how did these influence your decisions?

c) What risks are associated with not investigating assumptions during prototyping and testing?

Consider: Iterative development, prototyping considerations, risk assessment

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a) We plan to use the prototyping results to decide which parts of our concept should be kept, changed, or developed further. We mainly focused on testing four important assumptions: available rooftop area, residents' willingness to participate, whether the combined system can meet local energy demand, and whether the proposed storage capacity is enough.

The results will directly influence our next iteration. For example, if there is not enough suitable rooftop area, we can reduce the role of rooftop PV and rely more on a centralized PV park. If residents are not willing to participate, we can investigate the reasons, consider incentives, or use more public and commercial rooftops. If the energy production or storage capacity is not enough, we can resize the system or change the combination of technologies.

This process also helped us prioritize the assumptions that could have the biggest impact on our concept instead of trying to test everything at the same time.

b) When planning our prototype, we considered which assumptions were the most important and what we could realistically test with the available time, data, and resources. Since our concept is a large energy system, building a real physical prototype would not be practical. Therefore, we focused more on calculations, data collection, surveys, and simplified models.

We also considered what kind of data we needed and which stakeholders were relevant to include in the testing. For example, residents are relevant when testing willingness to participate, while BEOF and energy experts could be involved when testing energy demand, production, and storage capacity. These considerations helped us choose testing methods that were realistic and directly connected to our main assumptions.

c) If we do not investigate our assumptions, there is a risk that we continue developing a concept based on incorrect information. For example, we may assume that there is enough rooftop area or that enough residents are willing to participate, but this may not be true in reality.

There are also technical risks. If the solar system cannot produce enough energy, or if the battery and heat storage are too small, the system may not meet the local electricity and heat demand. This could lead to an unrealistic system design, higher costs, or major changes later in the project.

Therefore, testing the most important assumptions early can help us find problems before spending too much time and resources developing the concept.

